COL774 Assignment 3

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Decision Trees (and Random Forests)

Decision Tree Construction

Experimenting with Max Depth:

Max Depth	Train Accuracy	Validation Accuracy	Test Accuracy
5	0.8987	0.5885	0.5832
10	0.9974	0.5954	0.5874
15	1.0000	0.5954	0.5915
20	1.0000	0.5954	0.5915

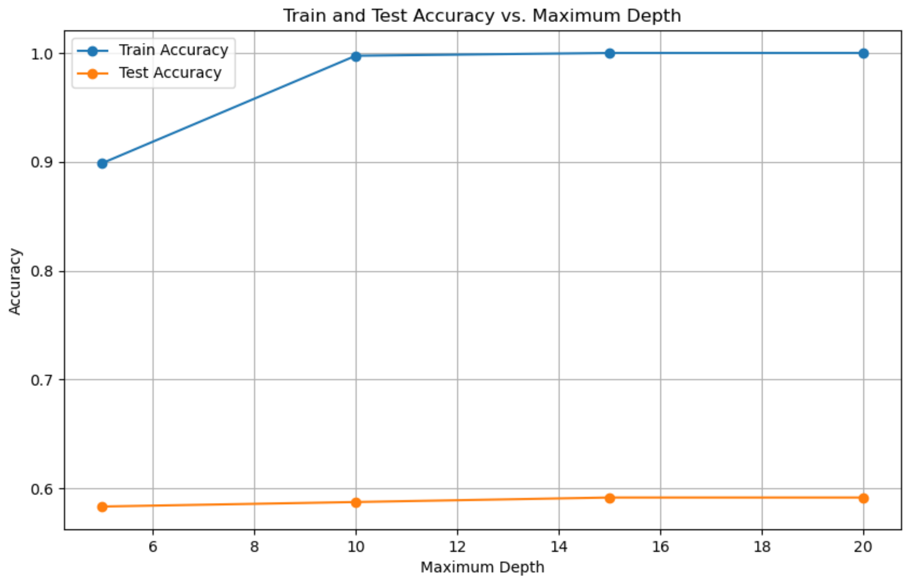

Figure 1: Accuracy vs Max Depth Plot

We observe **severe overfitting**. Although the training accuracy rapidly climbs to 100% as depth increases, the validation and test accuracy plateau at a very low 59.5% and 59.1%, respectively. This wide gap indicates that the model is simply memorizing the training data rather than learning generalizable patterns, and increasing the depth beyond 10 does not provide a meaningful benefit to its predictive power on unseen data.

Decision Tree One Hot Encoding

Experimenting with Max Depth:

Max Depth	Train Accuracy	Validation Accuracy	Test Accuracy
15	0.7350	0.5782	0.5677
25	0.8588	0.6000	0.5843
35	0.9554	0.6000	0.6029
45	0.9992	0.5943	0.6029

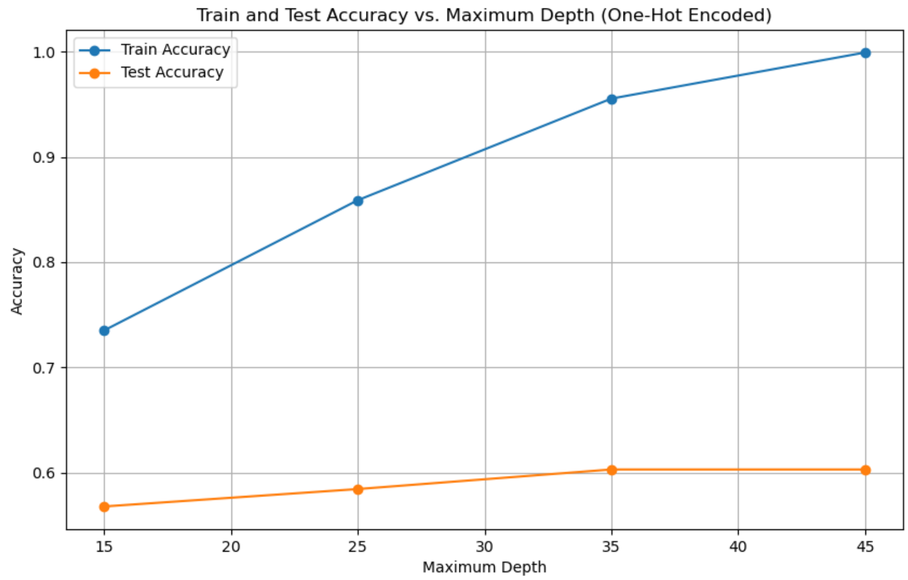

Figure 2: Accuracy vs Max Depth plot

Just like the last one, this model is **severely overfitting**, as the training accuracy shoots up to 100% while the test accuracy barely moves. However, this one-hot encoded version is marginally better, peaking at 60.3% test accuracy compared to the previous 59.2%. It also seems to **overfit more slowly**, needing a much deeper tree to memorize the training data, likely because one-hot encoding created **more features**.

Decision Tree Post Pruning

Initial Tree Performance (Pre-Pruning):

Initial Max Depth	Nodes	Train Accuracy	Val Accuracy	Test Accuracy
15	1417	0.7350	0.5782	0.5677
25	2499	0.8588	0.6000	0.5843
35	3403	0.9554	0.6000	0.6029
45	3953	0.9992	0.5943	0.6029

Final Pruned Tree Performance (Post-Pruning):

Initial Max Depth	Pruned Nodes	Train Accuracy	Val Accuracy	Test Accuracy
15	781	0.6720	0.6414	0.5646
25	1337	0.7567	0.7000	0.5957
35	1885	0.8118	0.7345	0.6122
45	1995	0.8205	0.7345	0.6153

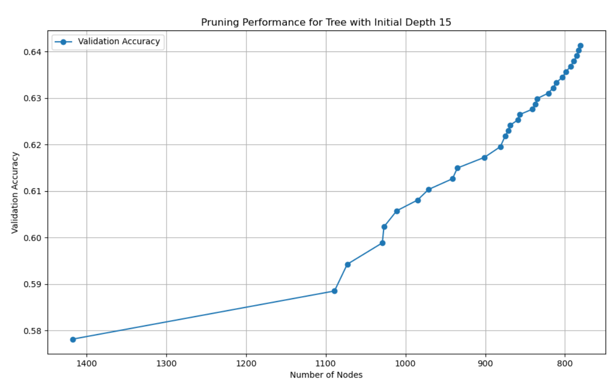

Figure 3: Max Depth = 15

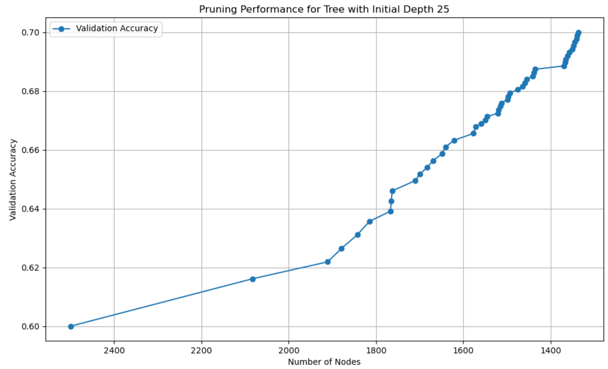

Figure 4: Max Depth = 25

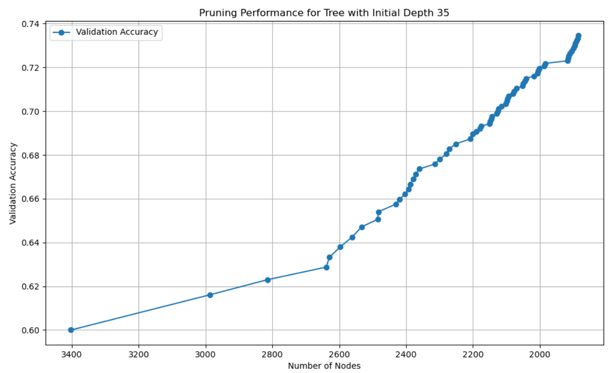

Figure 5: Max Depth = 35

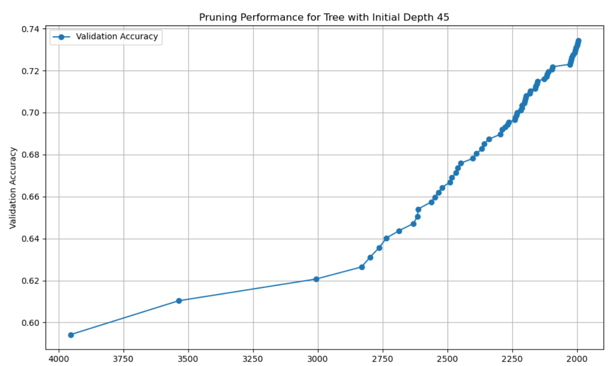

Figure 6: Max Depth = 45

Post-pruning proved to be highly effective, improving validation accuracy from a low of 60% to a peak of 73.45%. This process works by significantly **reducing overfitting**; for example, it reduced the number of nodes in the depth 45 tree from 3953 to 1995. This reduction closes the large gap between training and validation accuracy and results in a **more generalized model**. The **best-performing model** was achieved by **first growing a very deep overfit tree (like max depth 45) and then pruning it back**, which yielded the highest test accuracy so far at 61.53%. The plots confirm this, showing that the validation accuracy increases as the overfit nodes are removed.

Pruning with Gini Impurity

Initial Gini Tree Performance (Pre-Pruning):

Initial Max Depth	Nodes	Validation Accuracy
15	1623	0.5977
25	3229	0.6218
35	3923	0.6138
45	4029	0.6161

Final Pruned Gini Tree Performance (Post-Pruning):

Initial Max Depth	Final Validation Accuracy	Final Test Accuracy
15	0.6793	0.5739
25	0.7368	0.6050
35	0.7598	0.6019
45	0.7598	0.5998

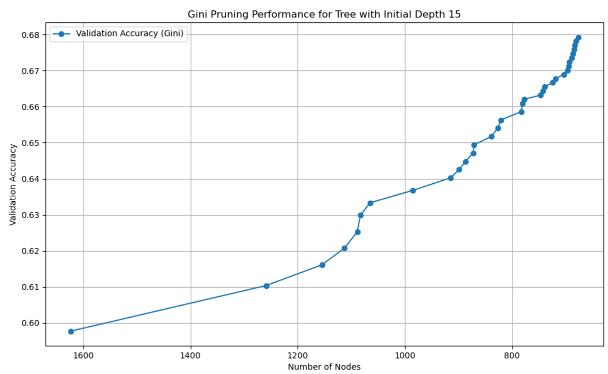

Figure 7: Max Depth = 15

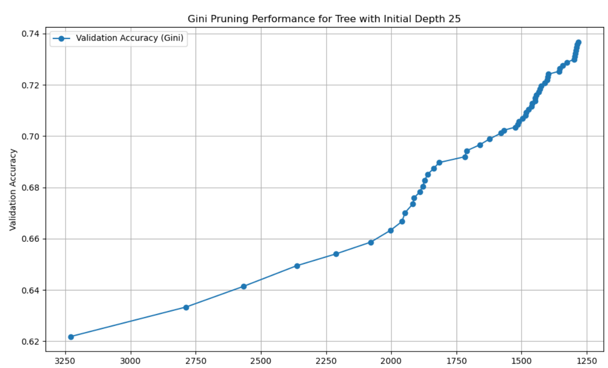

Figure 8: Max Depth = 25

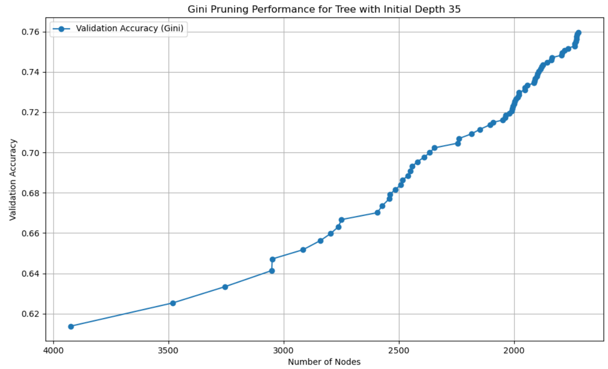

Figure 9: Max Depth = 35

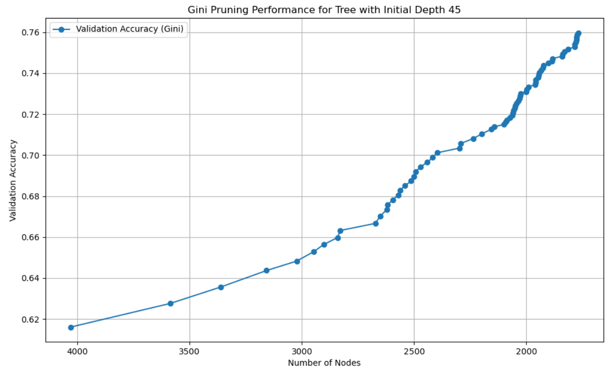

Figure 10: Max Depth = 45

Pruning was once again very effective, **increasing the accuracy of validation** from a low of 61-62% to a new peak of 76% (0.7598). This suggests that the Gini criterion was able to find a model that was **even better generalized** to the validation set.

However, this higher validation accuracy **did not translate into a better test score**. The Gini model's best test accuracy was 60.5%, which is slightly worse than the 61.5% achieved by the pruned entropy model. The plots all confirm the same successful pruning process, showing **increased accuracy as the overfit nodes are removed**.

Decision Tree sci-kit learn

Varying **max depth**:

Max Depth	Train Accuracy	Validation Accuracy	Test Accuracy
15	0.7267	0.5920	0.6039
25	0.8615	0.6069	0.6153
35	0.9599	0.5931	0.6194
45	0.9953	0.6034	0.6194

Varying **ccp alpha**:

ccp_alpha	Train Accuracy	Validation Accuracy	Test Accuracy
0.0	1.0000	0.6138	0.6153
0.0001	1.0000	0.6138	0.6153
0.0003	0.9761	0.6092	0.6194
0.0005	0.8471	0.6195	0.6143

Model Summary:

Tuning Method	Best Parameter	Validation Accuracy	Test Accuracy
Depth Tuning	depth=25	0.6069	0.6153
Alpha (Pruning) Tuning	alpha=0.0005	0.6195	0.6143

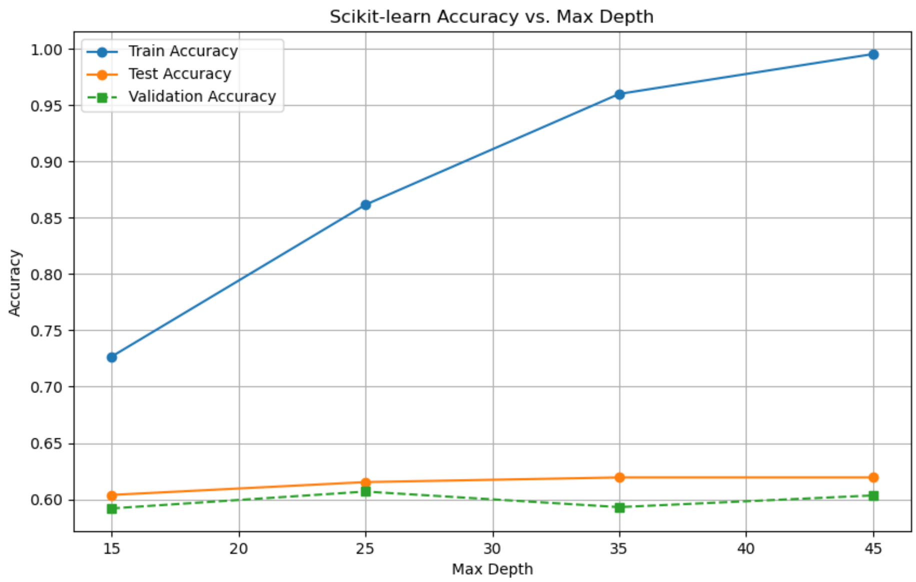

Figure 11: Accuracy vs Max Depth plot

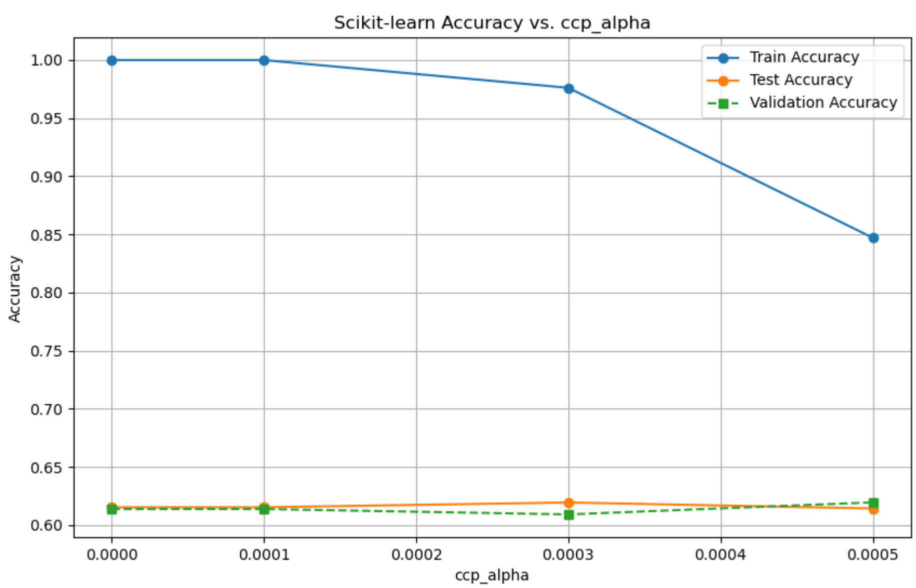

Figure 12: Accuracy vs alpha plot

The scikit-learn model also shows **severe overfitting** when tuning max depth, as seen in the large gap between the training and test accuracy plots. The more effective strategy was tuning ccp alpha, where pruning controlled this overfitting and found a better model. The best scikit-learn model, chosen by alpha=0.0005, achieved a peak validation accuracy of 0.6195 and a test accuracy of 0.6143. This performance is **not significantly different from the best from-scratch pruned model**, indicating that both implementations performed comparably on this dataset.

Random Forests

Best Parameters Found:

Parameter	**Value**
max_features	0.9
min_samples_split	10
n_estimators	250

Optimal Model Performance:

Metric	**Accuracy**
Training Accuracy	0.9848
OOB Accuracy	0.7162
Validation Accuracy	0.7115
Test Accuracy	0.6949

The Random Forest model provides a **performance jump**. Its final test accuracy of 0.6949 beats the best results from the pruned single-tree models (0.6153 for Entropy and 0.6050 for Gini). This shows that for this dataset, ensembling (bagging) is a **more effective strategy** than pruning a single tree. While the training accuracy (0.9848) is still high, the **model generalizes much better**, which is confirmed by the strong OOB (0.7162) and Validation (0.7115) scores. The fact that the OOB and validation scores are so close also indicates that the grid search found a **reliable model**.

Neural Networks

The stopping criteria used in parts b,c,d,e and f: We employ an **Early Stopping** criterion to prevent overfitting and ensure a fair comparison between architectures. The training data is split, reserving 10% as an internal validation set. We monitor the macro F1-score on this validation set after each epoch. If the score does not improve for a patience of 10 consecutive epochs, training is halted, and the model weights from the best-performing epoch are restored.
A maximum epoch limit of 375 is set as a safeguard.

Part (b)

Table 1: Comparison of different hidden layer units

Hidden Units	Train P	Train R	Train F1	Test P	Test R	Test F1
1	0.0087	0.0456	0.0129	0.0090	0.0465	0.0139
5	0.3210	0.3419	0.3079	0.2984	0.3231	0.2907
10	0.5742	0.5806	0.5732	0.5506	0.5562	0.5487
50	0.8253	0.8253	0.8250	0.7546	0.7539	0.7534
100	0.9028	0.9028	0.9027	0.8138	0.8134	0.8131

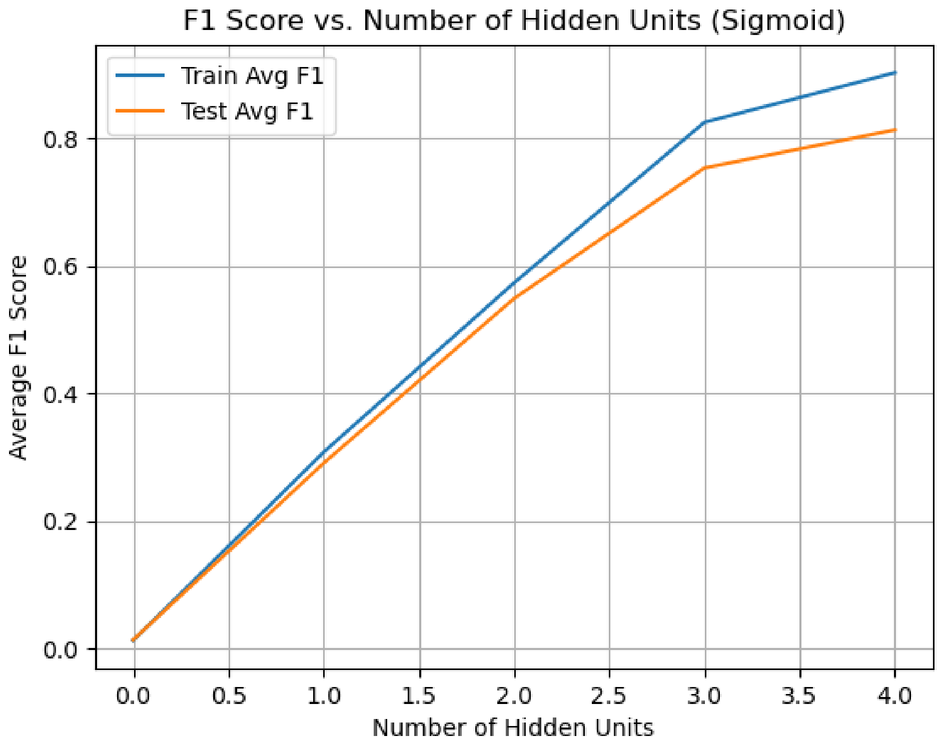

Figure 13: F1 score vs Number of hidden units (Sigmoid)

The results clearly show that model capacity is crucial, as **performance scales directly with the number of hidden units**. The models with 1 and 5 units are severely underfit, achieving almost no predictive power. Performance improves with 50 and 100 units, peaking at a Test F1 score of 81.31% for the 100-unit model. At this point, a gap emerges between the Train F1 (90.3%) and Test F1 (81.3%), indicating the model is beginning to overfit, though it still generalizes far better than the simpler models.

Table 2: Classification Reports for h=1

(a) Train (h=1)

Class	P	R	F1	Support
0	0.00	0.00	0.00	600
1	0.00	0.00	0.00	600
2	0.00	0.00	0.00	600
3	0.00	0.00	0.00	600
4	0.00	0.00	0.00	600
5	0.00	0.00	0.00	600
6	0.00	0.00	0.00	600
7	0.03	0.14	0.05	600
8	0.00	0.00	0.00	600
9	0.00	0.00	0.00	600
10	0.04	0.06	0.05	600
11	0.00	0.00	0.00	600
12	0.00	0.00	0.00	600
13	0.00	0.00	0.00	600
14	0.05	0.20	0.07	600
15	0.00	0.00	0.00	600
16	0.00	0.00	0.00	600
17	0.03	0.41	0.06	600
18	0.00	0.00	0.00	600
19	0.04	0.04	0.04	600
20	0.00	0.00	0.00	600
21	0.00	0.00	0.00	600
22	0.00	0.00	0.00	600
23	0.08	0.38	0.13	600
24	0.00	0.00	0.00	600
25	0.00	0.00	0.00	600
26	0.00	0.00	0.00	600
27	0.02	0.04	0.03	600
28	0.09	0.48	0.14	600
29	0.00	0.00	0.00	600
30	0.00	0.00	0.00	600
31	0.00	0.00	0.00	600
32	0.00	0.00	0.00	600
33	0.00	0.00	0.00	600
34	0.00	0.00	0.00	600
35	0.00	0.00	0.00	600
Accuracy			0.05	21600
Macro Avg	0.01	0.05	0.02	21600
Weighted Avg	0.01	0.05	0.02	21600

(b) Test (h=1)

Class	P	R	F1	Support
0	0.00	0.00	0.00	300
1	0.00	0.00	0.00	300
2	0.00	0.00	0.00	300
3	0.00	0.00	0.00	300
4	0.00	0.00	0.00	300
5	0.00	0.00	0.00	300
6	0.00	0.00	0.00	300
7	0.03	0.12	0.04	300
8	0.00	0.00	0.00	300
9	0.00	0.00	0.00	300
10	0.05	0.07	0.06	300
11	0.00	0.00	0.00	300
12	0.00	0.00	0.00	300
13	0.00	0.00	0.00	300
14	0.06	0.28	0.10	300
15	0.00	0.00	0.00	300
16	0.00	0.00	0.00	300
17	0.03	0.38	0.06	300
18	0.00	0.00	0.00	300
19	0.03	0.03	0.03	300
20	0.00	0.00	0.00	300
21	0.00	0.00	0.00	300
22	0.00	0.00	0.00	300
23	0.08	0.38	0.13	300
24	0.00	0.00	0.00	300
25	0.00	0.00	0.00	300
26	0.00	0.00	0.00	300
27	0.03	0.04	0.03	300
28	0.08	0.48	0.14	300
29	0.00	0.00	0.00	300
30	0.00	0.00	0.00	300
31	0.00	0.00	0.00	300
32	0.00	0.00	0.00	300
33	0.00	0.00	0.00	300
34	0.00	0.00	0.00	300
35	0.00	0.00	0.00	300
Accuracy			0.05	10800
Macro Avg	0.01	0.05	0.02	10800
Weighted Avg	0.01	0.05	0.02	10800

Table 3: Classification Reports for h=5

(a) Train (h=5)

Class	P	R	F1	Support
0	0.45	0.77	0.57	600
1	0.13	0.01	0.03	600
2	0.24	0.23	0.23	600
3	0.28	0.15	0.19	600
4	0.17	0.03	0.04	600
5	0.46	0.60	0.52	600
6	0.32	0.29	0.30	600
7	0.45	0.56	0.50	600
8	0.46	0.57	0.51	600
9	0.55	0.68	0.61	600
10	0.50	0.73	0.60	600
11	0.59	0.42	0.49	600
12	0.42	0.62	0.50	600
13	0.45	0.67	0.54	600
14	0.29	0.52	0.37	600
15	0.44	0.69	0.54	600
16	0.28	0.18	0.22	600
17	0.37	0.49	0.42	600
18	0.37	0.50	0.43	600
19	0.39	0.41	0.40	600
20	0.34	0.44	0.38	600
21	0.43	0.77	0.56	600
22	0.38	0.14	0.21	600
23	0.45	0.50	0.47	600
24	0.30	0.32	0.31	600
25	0.18	0.05	0.08	600
26	0.43	0.44	0.44	600
27	0.45	0.39	0.42	600
28	0.29	0.06	0.09	600
29	0.32	0.31	0.31	600
30	0.35	0.27	0.31	600
31	0.35	0.40	0.38	600
32	0.41	0.40	0.41	600
33	0.38	0.20	0.26	600
34	0.47	0.18	0.26	600
35	0.37	0.38	0.37	600
Accuracy			0.40	21600
Macro Avg	0.38	0.40	0.37	21600
Weighted Avg	0.38	0.40	0.37	21600

(b) Test (h=5)

Class	P	R	F1	Support
0	0.41	0.70	0.52	300
1	0.21	0.02	0.04	300
2	0.25	0.24	0.24	300
3	0.28	0.17	0.21	300
4	0.15	0.02	0.04	300
5	0.43	0.56	0.49	300
6	0.28	0.25	0.26	300
7	0.42	0.55	0.48	300
8	0.43	0.52	0.47	300
9	0.50	0.68	0.58	300
10	0.52	0.74	0.61	300
11	0.56	0.43	0.48	300
12	0.38	0.60	0.46	300
13	0.42	0.60	0.49	300
14	0.27	0.46	0.34	300
15	0.40	0.67	0.50	300
16	0.28	0.18	0.22	300
17	0.37	0.46	0.41	300
18	0.35	0.47	0.40	300
19	0.38	0.35	0.36	300
20	0.35	0.40	0.37	300
21	0.42	0.75	0.54	300
22	0.29	0.13	0.18	300
23	0.40	0.47	0.43	300
24	0.23	0.26	0.24	300
25	0.21	0.06	0.10	300
26	0.40	0.43	0.41	300
27	0.44	0.36	0.40	300
28	0.40	0.07	0.12	300
29	0.35	0.33	0.34	300
30	0.26	0.20	0.23	300
31	0.34	0.41	0.37	300
32	0.38	0.33	0.35	300
33	0.30	0.14	0.20	300
34	0.44	0.17	0.25	300
35	0.38	0.37	0.37	300
Accuracy			0.38	10800
Macro Avg	0.36	0.38	0.35	10800
Weighted Avg	0.36	0.38	0.35	10800

Table 4: Classification Reports for h=10

(a) Train (h=10)

Class	P	R	F1	Support
0	0.67	0.76	0.71	600
1	0.41	0.24	0.30	600
2	0.54	0.49	0.52	600
3	0.47	0.48	0.48	600
4	0.54	0.43	0.48	600
5	0.60	0.66	0.63	600
6	0.49	0.45	0.47	600
7	0.65	0.68	0.66	600
8	0.76	0.77	0.77	600
9	0.67	0.68	0.68	600
10	0.64	0.74	0.69	600
11	0.62	0.52	0.57	600
12	0.56	0.56	0.56	600
13	0.57	0.76	0.65	600
14	0.55	0.65	0.59	600
15	0.57	0.73	0.64	600
16	0.48	0.43	0.45	600
17	0.46	0.39	0.42	600
18	0.57	0.54	0.55	600
19	0.61	0.67	0.64	600
20	0.47	0.60	0.52	600
21	0.69	0.76	0.72	600
22	0.65	0.46	0.54	600
23	0.66	0.59	0.62	600
24	0.55	0.36	0.44	600
25	0.34	0.23	0.27	600
26	0.63	0.75	0.68	600
27	0.62	0.61	0.62	600
28	0.51	0.43	0.47	600
29	0.47	0.56	0.51	600
30	0.57	0.65	0.61	600
31	0.44	0.42	0.43	600
32	0.51	0.55	0.53	600
33	0.60	0.70	0.64	600
34	0.62	0.54	0.58	600
35	0.58	0.65	0.61	600
Accuracy			0.57	21600
Macro Avg	0.56	0.57	0.56	21600
Weighted Avg	0.56	0.57	0.56	21600

(b) Test (h=10)

Class	P	R	F1	Support
0	0.61	0.71	0.66	300
1	0.39	0.23	0.29	300
2	0.53	0.51	0.52	300
3	0.47	0.47	0.47	300
4	0.51	0.43	0.47	300
5	0.57	0.64	0.60	300
6	0.45	0.40	0.42	300
7	0.59	0.64	0.61	300
8	0.72	0.81	0.76	300
9	0.70	0.68	0.69	300
10	0.66	0.75	0.70	300
11	0.58	0.49	0.53	300
12	0.55	0.50	0.53	300
13	0.53	0.73	0.62	300
14	0.50	0.56	0.53	300
15	0.52	0.72	0.60	300
16	0.43	0.43	0.43	300
17	0.50	0.40	0.44	300
18	0.56	0.55	0.56	300
19	0.56	0.57	0.56	300
20	0.45	0.57	0.51	300
21	0.68	0.72	0.70	300
22	0.58	0.42	0.49	300
23	0.66	0.58	0.61	300
24	0.52	0.33	0.40	300
25	0.33	0.21	0.26	300
26	0.59	0.73	0.65	300
27	0.57	0.52	0.55	300
28	0.46	0.36	0.40	300
29	0.47	0.58	0.52	300
30	0.56	0.64	0.60	300
31	0.40	0.42	0.41	300
32	0.51	0.56	0.53	300
33	0.60	0.65	0.62	300
34	0.57	0.50	0.53	300
35	0.55	0.60	0.57	300
Accuracy			0.54	10800
Macro Avg	0.54	0.54	0.54	10800
Weighted Avg	0.54	0.54	0.54	10800

Table 5: Classification Reports for h=50

(a) Train (h=50)

Class	P	R	F1	Support
0	0.89	0.86	0.88	600
1	0.73	0.79	0.76	600
2	0.82	0.82	0.82	600
3	0.71	0.71	0.71	600
4	0.72	0.76	0.74	600
5	0.85	0.83	0.84	600
6	0.82	0.77	0.79	600
7	0.87	0.85	0.86	600
8	0.91	0.91	0.91	600
9	0.85	0.86	0.86	600
10	0.89	0.90	0.89	600
11	0.87	0.87	0.87	600
12	0.78	0.76	0.77	600
13	0.87	0.86	0.87	600
14	0.78	0.82	0.80	600
15	0.85	0.86	0.86	600
16	0.71	0.69	0.70	600
17	0.81	0.73	0.77	600
18	0.74	0.76	0.75	600
19	0.84	0.85	0.85	600
20	0.71	0.79	0.75	600
21	0.87	0.89	0.88	600
22	0.81	0.74	0.77	600
23	0.80	0.80	0.80	600
24	0.78	0.71	0.74	600
25	0.65	0.59	0.62	600
26	0.87	0.88	0.87	600
27	0.87	0.88	0.87	600
28	0.70	0.70	0.70	600
29	0.78	0.83	0.80	600
30	0.77	0.80	0.80	600
31	0.72	0.68	0.70	600
32	0.78	0.79	0.78	600
33	0.85	0.85	0.85	600
34	0.73	0.74	0.74	600
35	0.83	0.88	0.85	600
Accuracy			0.80	21600
Macro Avg	0.80	0.80	0.80	21600
Weighted Avg	0.80	0.80	0.80	21600

(b) Test (h=50)

Class	P	R	F1	Support
0	0.83	0.84	0.83	300
1	0.65	0.70	0.67	300
2	0.76	0.72	0.74	300
3	0.62	0.67	0.64	300
4	0.66	0.68	0.67	300
5	0.80	0.76	0.78	300
6	0.73	0.68	0.70	300
7	0.80	0.82	0.81	300
8	0.85	0.86	0.85	300
9	0.86	0.85	0.86	300
10	0.91	0.87	0.89	300
11	0.81	0.83	0.82	300
12	0.73	0.71	0.72	300
13	0.79	0.83	0.81	300
14	0.75	0.79	0.77	300
15	0.77	0.85	0.81	300
16	0.64	0.64	0.64	300
17	0.71	0.65	0.68	300
18	0.68	0.71	0.70	300
19	0.72	0.72	0.72	300
20	0.66	0.69	0.67	300
21	0.83	0.84	0.84	300
22	0.73	0.64	0.68	300
23	0.76	0.73	0.74	300
24	0.69	0.68	0.68	300
25	0.61	0.53	0.57	300
26	0.81	0.84	0.82	300
27	0.86	0.81	0.83	300
28	0.65	0.59	0.62	300
29	0.73	0.78	0.75	300
30	0.71	0.80	0.75	300
31	0.62	0.61	0.61	300
32	0.73	0.70	0.71	300
33	0.76	0.81	0.78	300
34	0.68	0.69	0.68	300
35	0.79	0.79	0.79	300
Accuracy			0.74	10800
Macro Avg	0.74	0.74	0.74	10800
Weighted Avg	0.74	0.74	0.74	10800

Table 6: Classification Reports for h=100

(a) Train (h=100)

Class	P	R	F1	Support
0	0.91	0.92	0.91	600
1	0.82	0.82	0.82	600
2	0.80	0.81	0.81	600
3	0.82	0.79	0.81	600
4	0.86	0.86	0.86	600
5	0.83	0.79	0.81	600
6	0.89	0.87	0.88	600
7	0.94	0.93	0.93	600
8	0.89	0.88	0.88	600
9	0.91	0.94	0.92	600
10	0.92	0.89	0.90	600
11	0.83	0.83	0.83	600
12	0.90	0.90	0.90	600
13	0.85	0.86	0.85	600
14	0.89	0.89	0.89	600
15	0.77	0.78	0.77	600
16	0.84	0.84	0.84	600
17	0.83	0.82	0.82	600
18	0.88	0.86	0.87	600
19	0.80	0.89	0.84	600
20	0.91	0.93	0.92	600
21	0.83	0.78	0.81	600
22	0.86	0.84	0.85	600
23	0.83	0.81	0.81	600
24	0.78	0.75	0.76	600
25	0.91	0.91	0.91	600
26	0.90	0.92	0.91	600
27	0.78	0.77	0.77	600
28	0.85	0.86	0.86	600
29	0.84	0.90	0.87	600
30	0.78	0.76	0.77	600
31	0.86	0.85	0.85	600
32	0.89	0.89	0.89	600
33	0.77	0.82	0.79	600
34	0.85	0.90	0.88	600
35	0.85	0.85	0.85	600
Accuracy			0.85	21600
Macro Avg	0.85	0.85	0.85	21600
Weighted Avg	0.85	0.85	0.85	21600

(b) Test (h=100)

Class	P	R	F1	Support
0	0.88	0.89	0.89	300
1	0.68	0.72	0.70	300
2	0.82	0.73	0.77	300
3	0.69	0.70	0.70	300
4	0.76	0.71	0.73	300
5	0.81	0.81	0.81	300
6	0.75	0.72	0.74	300
7	0.82	0.83	0.83	300
8	0.87	0.89	0.88	300
9	0.88	0.86	0.87	300
10	0.94	0.92	0.93	300
11	0.85	0.86	0.86	300
12	0.77	0.78	0.77	300
13	0.80	0.87	0.83	300
14	0.79	0.83	0.81	300
15	0.86	0.87	0.86	300
16	0.67	0.70	0.69	300
17	0.78	0.73	0.75	300
18	0.76	0.76	0.76	300
19	0.75	0.75	0.75	300
20	0.73	0.80	0.76	300
21	0.85	0.88	0.87	300
22	0.76	0.68	0.72	300
23	0.81	0.76	0.78	300
24	0.73	0.72	0.73	300
25	0.69	0.66	0.67	300
26	0.85	0.85	0.85	300
27	0.86	0.84	0.85	300
28	0.72	0.69	0.71	300
29	0.79	0.82	0.81	300
30	0.75	0.82	0.78	300
31	0.69	0.67	0.68	300
32	0.76	0.74	0.75	300
33	0.80	0.82	0.81	300
34	0.72	0.77	0.74	300
35	0.82	0.80	0.81	300
Accuracy			0.79	10800
Macro Avg	0.79	0.79	0.78	10800
Weighted Avg	0.79	0.79	0.78	10800

Part (c)

Table 7: Precision, Recall, F1 comparison for different Network depths

Architecture	Train P	Train R	Train F1	Test P	Test R	Test F1
[512]	0.8683	0.8679	0.8678	0.8026	0.8017	0.8013
[512, 256]	0.9487	0.9486	0.9484	0.8625	0.8620	0.8620
[512, 256, 128]	0.9612	0.9611	0.9611	0.8467	0.8459	0.8459
[512, 256, 128, 64]	0.0025	0.0325	0.0042	0.0024	0.0318	0.0039

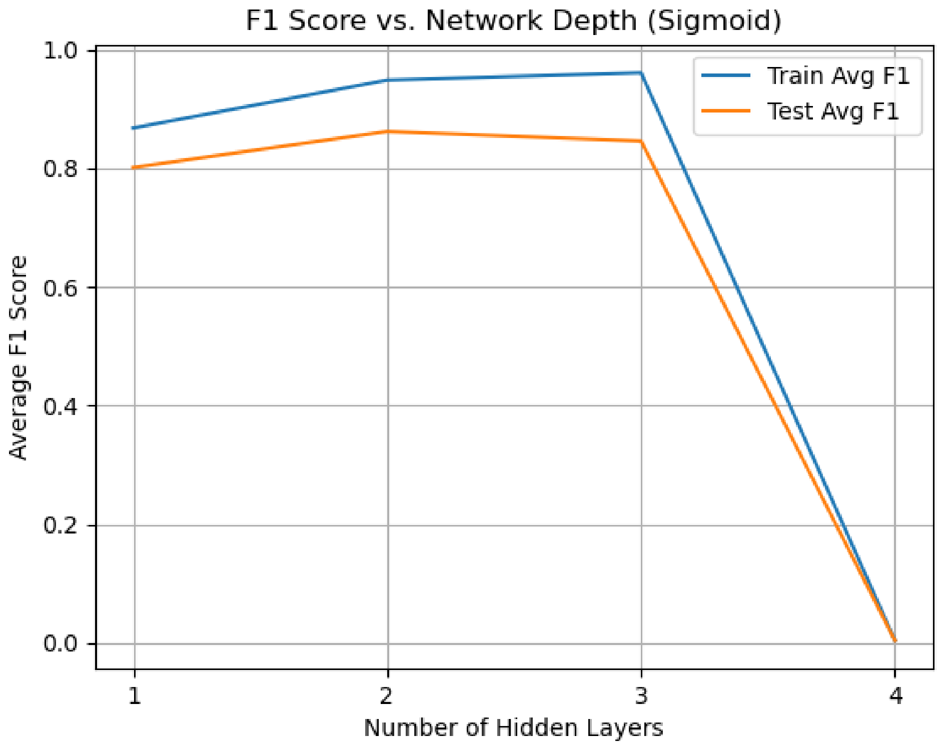

Figure 14: F1 Score vs Network Depth (Sigmoid)

The results show that **network depth is a critical factor**. Performance peaked with the 2-layer [512, 256] network, which achieved the highest Test F1 score of 86.2%. However, adding a third layer slightly degraded performance, and the 4-layer network experienced a catastrophic training failure, with its F1 score dropping to near zero. This is a demonstration of the **vanishing gradient problem**, where the Sigmoid activation's small derivatives multiply to zero during backpropagation, preventing the deeper network from learning.

Part (d)

Table 8: Precision, Recall, F1 comparison for different Network depths

Architecture	Train P	Train R	Train F1	Test P	Test R	Test F1
[512]	0.9816	0.9816	0.9816	0.8657	0.8653	0.8650
[512, 256]	0.9860	0.9860	0.9860	0.8751	0.8745	0.8745
[512, 256, 128]	0.9888	0.9888	0.9887	0.8763	0.8759	0.8758
[512, 256, 128, 64]	0.9860	0.9859	0.9859	0.8750	0.8741	0.8741

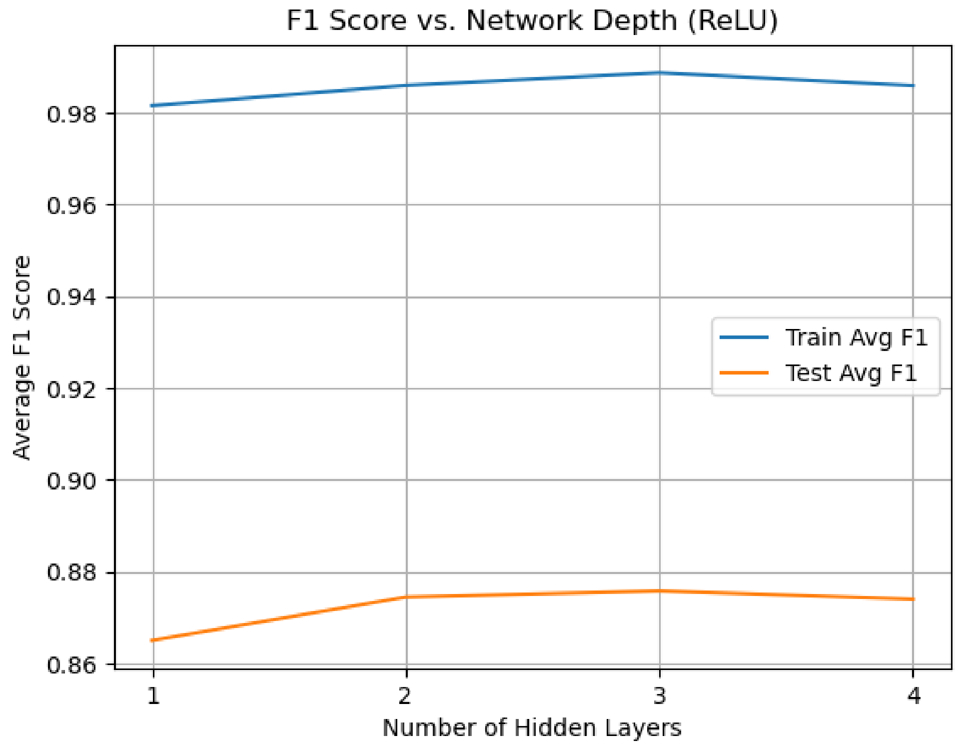

Figure 15: F1 Score vs Network Depth (ReLU)

Switching to the ReLU activation function **solves the vanishing gradient problem** observed with Sigmoid. The plot shows that all architectures, including the 4-layer network, now train successfully, with test F1 scores remaining stable and high. Performance peaked with the 3-layer [512, 256, 128] network, which achieved a Test F1 score of 87.6%, slightly outperforming the best Sigmoid model. Although a gap between the 98% training F1 and 87% test F1 indicates some overfitting, ReLU enables the deeper networks to learn effectively and generalize well.

Table 9: Classification Reports for Architecture [512]

(a) Train [512]

Class	P	R	F1	Support
0	0.86	0.87	0.86	600
1	0.86	0.84	0.85	600
2	0.89	0.84	0.86	600
3	0.87	0.84	0.86	600
4	0.90	0.93	0.92	600
5	0.85	0.87	0.86	600
6	0.87	0.84	0.86	600
7	0.92	0.91	0.92	600
8	0.94	0.94	0.94	600
9	0.92	0.93	0.92	600
10	0.91	0.94	0.93	600
11	0.90	0.90	0.90	600
12	0.86	0.89	0.87	600
13	0.91	0.93	0.92	600
14	0.91	0.91	0.91	600
15	0.92	0.91	0.92	600
16	0.85	0.86	0.85	600
17	0.89	0.85	0.87	600
18	0.88	0.88	0.88	600
19	0.89	0.91	0.90	600
20	0.82	0.85	0.83	600
21	0.94	0.94	0.94	600
22	0.86	0.81	0.83	600
23	0.90	0.90	0.90	600
24	0.84	0.83	0.83	600
25	0.77	0.74	0.75	600
26	0.90	0.91	0.91	600
27	0.90	0.93	0.91	600
28	0.81	0.83	0.82	600
29	0.85	0.88	0.86	600
30	0.84	0.84	0.84	600
31	0.80	0.82	0.81	600
32	0.85	0.86	0.86	600
33	0.93	0.92	0.92	600
34	0.85	0.86	0.86	600
35	0.85	0.86	0.85	600
Accuracy			0.88	21600
Macro Avg	0.88	0.88	0.88	21600
Weighted Avg	0.88	0.88	0.88	21600

(b) Test [512]

Class	P	R	F1	Support
0	0.85	0.85	0.85	300
1	0.72	0.74	0.73	300
2	0.80	0.74	0.77	300
3	0.77	0.73	0.75	300
4	0.85	0.85	0.85	300
5	0.83	0.85	0.84	300
6	0.80	0.73	0.76	300
7	0.85	0.86	0.85	300
8	0.89	0.91	0.90	300
9	0.88	0.90	0.89	300
10	0.88	0.91	0.89	300
11	0.84	0.85	0.85	300
12	0.79	0.83	0.81	300
13	0.85	0.89	0.87	300
14	0.86	0.85	0.86	300
15	0.88	0.89	0.88	300
16	0.77	0.80	0.78	300
17	0.83	0.78	0.81	300
18	0.81	0.82	0.82	300
19	0.80	0.84	0.82	300
20	0.73	0.75	0.74	300
21	0.89	0.90	0.89	300
22	0.79	0.73	0.76	300
23	0.85	0.84	0.85	300
24	0.76	0.77	0.77	300
25	0.69	0.65	0.67	300
26	0.86	0.86	0.86	300
27	0.85	0.89	0.87	300
28	0.75	0.78	0.76	300
29	0.80	0.82	0.81	300
30	0.80	0.79	0.79	300
31	0.70	0.75	0.72	300
32	0.78	0.80	0.79	300
33	0.89	0.88	0.89	300
34	0.80	0.81	0.81	300
35	0.84	0.87	0.85	300
Accuracy			0.81	10800
Macro Avg	0.81	0.81	0.81	10800
Weighted Avg	0.81	0.81	0.81	10800

Table 10: Classification Reports for Architecture [512, 256]

(a) Train [512, 256]

Class	P	R	F1	Support
0	0.94	0.94	0.94	600
1	0.94	0.94	0.94	600
2	0.96	0.94	0.95	600
3	0.94	0.92	0.93	600
4	0.95	0.97	0.96	600
5	0.95	0.95	0.95	600
6	0.96	0.94	0.95	600
7	0.97	0.98	0.97	600
8	0.98	0.98	0.98	600
9	0.96	0.97	0.97	600
10	0.97	0.97	0.97	600
11	0.96	0.96	0.96	600
12	0.95	0.96	0.95	600
13	0.96	0.97	0.96	600
14	0.96	0.96	0.96	600
15	0.97	0.97	0.97	600
16	0.93	0.93	0.93	600
17	0.95	0.94	0.95	600
18	0.95	0.94	0.95	600
19	0.96	0.96	0.96	600
20	0.93	0.94	0.93	600
21	0.98	0.98	0.98	600
22	0.94	0.93	0.93	600
23	0.96	0.96	0.96	600
24	0.93	0.92	0.92	600
25	0.89	0.88	0.88	600
26	0.96	0.96	0.96	600
27	0.96	0.97	0.97	600
28	0.92	0.92	0.92	600
29	0.93	0.95	0.94	600
30	0.94	0.94	0.94	600
31	0.91	0.92	0.91	600
32	0.94	0.94	0.94	600
33	0.97	0.97	0.97	600
34	0.93	0.94	0.93	600
35	0.96	0.97	0.97	600
Accuracy			0.95	21600
Macro Avg	0.95	0.95	0.95	21600
Weighted Avg	0.95	0.95	0.95	21600

(b) Test [512, 256]

Class	P	R	F1	Support
0	0.87	0.88	0.87	300
1	0.82	0.80	0.81	300
2	0.88	0.84	0.86	300
3	0.82	0.82	0.82	300
4	0.90	0.90	0.90	300
5	0.88	0.90	0.89	300
6	0.87	0.82	0.84	300
7	0.90	0.91	0.91	300
8	0.93	0.95	0.94	300
9	0.90	0.93	0.92	300
10	0.92	0.93	0.93	300
11	0.88	0.89	0.89	300
12	0.87	0.88	0.87	300
13	0.89	0.92	0.91	300
14	0.90	0.89	0.89	300
15	0.91	0.93	0.92	300
16	0.84	0.85	0.84	300
17	0.87	0.84	0.85	300
18	0.88	0.86	0.87	300
19	0.88	0.90	0.89	300
20	0.82	0.83	0.83	300
21	0.93	0.94	0.93	300
22	0.83	0.81	0.82	300
23	0.88	0.89	0.88	300
24	0.83	0.82	0.82	300
25	0.77	0.74	0.75	300
26	0.91	0.91	0.91	300
27	0.91	0.92	0.91	300
28	0.83	0.83	0.83	300
29	0.86	0.88	0.87	300
30	0.87	0.87	0.87	300
31	0.81	0.83	0.82	300
32	0.86	0.87	0.87	300
33	0.93	0.92	0.93	300
34	0.85	0.85	0.85	300
35	0.86	0.87	0.87	300
Accuracy			0.87	10800
Macro Avg	0.87	0.87	0.87	10800
Weighted Avg	0.87	0.87	0.87	10800

Table 11: Classification Reports for Architecture [512, 256, 128]

(a) Train [512, 256, 128]

Class	P	R	F1	Support
0	0.94	0.95	0.94	600
1	0.94	0.94	0.94	600
2	0.95	0.94	0.95	600
3	0.93	0.92	0.93	600
4	0.96	0.97	0.96	600
5	0.95	0.95	0.95	600
6	0.95	0.94	0.94	600
7	0.97	0.98	0.97	600
8	0.98	0.98	0.98	600
9	0.97	0.98	0.97	600
10	0.97	0.97	0.97	600
11	0.96	0.96	0.96	600
12	0.95	0.95	0.95	600
13	0.97	0.97	0.97	600
14	0.96	0.96	0.96	600
15	0.97	0.97	0.97	600
16	0.94	0.93	0.94	600
17	0.95	0.93	0.94	600
18	0.95	0.94	0.95	600
19	0.96	0.96	0.96	600
20	0.93	0.94	0.94	600
21	0.97	0.98	0.98	600
22	0.93	0.92	0.93	600
23	0.96	0.96	0.96	600
24	0.92	0.91	0.91	600
25	0.89	0.88	0.88	600
26	0.96	0.96	0.96	600
27	0.96	0.97	0.97	600
28	0.92	0.92	0.92	600
29	0.93	0.94	0.94	600
30	0.94	0.94	0.94	600
31	0.91	0.91	0.91	600
32	0.94	0.94	0.94	600
33	0.97	0.97	0.97	600
34	0.93	0.94	0.93	600
35	0.95	0.97	0.96	600
Accuracy			0.95	21600
Macro Avg	0.95	0.95	0.95	21600
Weighted Avg	0.95	0.95	0.95	21600

(b) Test [512, 256, 128]

Class	P	R	F1	Support
0	0.86	0.82	0.84	300
1	0.81	0.80	0.81	300
2	0.86	0.82	0.84	300
3	0.79	0.79	0.79	300
4	0.87	0.89	0.88	300
5	0.88	0.88	0.88	300
6	0.84	0.81	0.82	300
7	0.89	0.90	0.89	300
8	0.93	0.94	0.93	300
9	0.91	0.93	0.92	300
10	0.92	0.92	0.92	300
11	0.87	0.88	0.88	300
12	0.86	0.87	0.87	300
13	0.89	0.91	0.90	300
14	0.89	0.89	0.89	300
15	0.91	0.92	0.91	300
16	0.83	0.83	0.83	300
17	0.85	0.81	0.83	300
18	0.87	0.86	0.87	300
19	0.88	0.89	0.88	300
20	0.81	0.83	0.82	300
21	0.92	0.93	0.93	300
22	0.82	0.79	0.80	300
23	0.87	0.87	0.87	300
24	0.80	0.78	0.79	300
25	0.75	0.71	0.73	300
26	0.90	0.90	0.90	300
27	0.90	0.91	0.91	300
28	0.81	0.81	0.81	300
29	0.84	0.85	0.85	300
30	0.86	0.86	0.86	300
31	0.79	0.80	0.80	300
32	0.85	0.86	0.85	300
33	0.92	0.92	0.92	300
34	0.84	0.84	0.84	300
35	0.84	0.87	0.86	300
Accuracy			0.84	10800
Macro Avg	0.84	0.84	0.84	10800
Weighted Avg	0.84	0.84	0.84	10800

Table 12: Classification Reports for Architecture [512, 256, 128, 64]

(a) Train [512, 256, 128, 64]

Class	P	R	F1	Support
0	0.00	0.00	0.00	600
1	0.00	0.00	0.00	600
2	0.00	0.00	0.00	600
3	0.00	0.00	0.00	600
4	0.00	0.00	0.00	600
5	0.00	0.00	0.00	600
6	0.00	0.00	0.00	600
7	0.03	0.14	0.04	600
8	0.00	0.00	0.00	600
9	0.00	0.00	0.00	600
10	0.04	0.07	0.05	600
11	0.00	0.00	0.00	600
12	0.00	0.00	0.00	600
13	0.00	0.00	0.00	600
14	0.05	0.20	0.08	600
15	0.00	0.00	0.00	600
16	0.00	0.00	0.00	600
17	0.03	0.41	0.06	600
18	0.00	0.00	0.00	600
19	0.03	0.03	0.03	600
20	0.00	0.00	0.00	600
21	0.00	0.00	0.00	600
22	0.00	0.00	0.00	600
23	0.08	0.38	0.13	600
24	0.00	0.00	0.00	600
25	0.00	0.00	0.00	600
26	0.00	0.00	0.00	600
27	0.03	0.04	0.03	600
28	0.08	0.48	0.14	600
29	0.00	0.00	0.00	600
30	0.00	0.00	0.00	600
31	0.00	0.00	0.00	600
32	0.00	0.00	0.00	600
33	0.00	0.00	0.00	600
34	0.00	0.00	0.00	600
35	0.00	0.00	0.00	600
Accuracy			0.05	21600
Macro Avg	0.01	0.04	0.01	21600
Weighted Avg	0.01	0.05	0.01	21600

(b) Test [512, 256, 128, 64]

Class	P	R	F1	Support
0	0.00	0.00	0.00	300
1	0.00	0.00	0.00	300
2	0.00	0.00	0.00	300
3	0.00	0.00	0.00	300
4	0.00	0.00	0.00	300
5	0.00	0.00	0.00	300
6	0.00	0.00	0.00	300
7	0.03	0.12	0.04	300
8	0.00	0.00	0.00	300
9	0.00	0.00	0.00	300
10	0.05	0.07	0.06	300
11	0.00	0.00	0.00	300
12	0.00	0.00	0.00	300
13	0.00	0.00	0.00	300
14	0.06	0.28	0.10	300
15	0.00	0.00	0.00	300
16	0.00	0.00	0.00	300
17	0.03	0.38	0.06	300
18	0.00	0.00	0.00	300
19	0.03	0.03	0.03	300
20	0.00	0.00	0.00	300
21	0.00	0.00	0.00	300
22	0.00	0.00	0.00	300
23	0.08	0.38	0.13	300
24	0.00	0.00	0.00	300
25	0.00	0.00	0.00	300
26	0.00	0.00	0.00	300
27	0.03	0.04	0.03	300
28	0.08	0.48	0.14	300
29	0.00	0.00	0.00	300
30	0.00	0.00	0.00	300
31	0.00	0.00	0.00	300
32	0.00	0.00	0.00	300
33	0.00	0.00	0.00	300
34	0.00	0.00	0.00	300
35	0.00	0.00	0.00	300
Accuracy			0.05	10800
Macro Avg	0.01	0.04	0.01	10800
Weighted Avg	0.01	0.05	0.01	10800

Part (e)

Table 13: Precision, Recall, F1 comparison for different Network depths

Architecture	Train P	Train R	Train F1	Test P	Test R	Test F1
(512,)	0.9905	0.9905	0.9905	0.8986	0.8983	0.8982
(512, 256)	0.9912	0.9912	0.9912	0.9088	0.9086	0.9085
(512, 256, 128)	0.9903	0.9903	0.9903	0.9188	0.9186	0.9186
(512, 256, 128, 64)	0.9852	0.9851	0.9851	0.8982	0.8963	0.8964

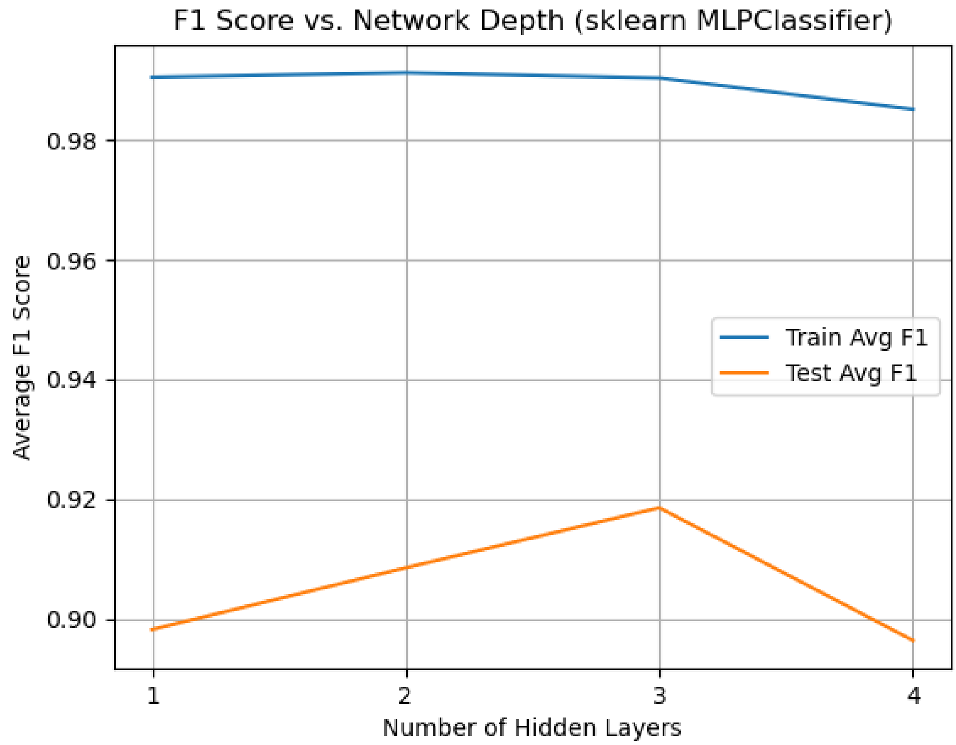

Figure 16: F1 Score vs Network Depth (sklearn MLPClassifier)

The scikit-learn MLPClassifier shows a performance improvement over our from-scratch ReLU implementation, achieving a new peak Test F1 score of 91.9% with the 3-layer [512, 256, 128] network. The plot clearly shows that model performance increases with depth, peaking at 3 layers, and then begins to degrade at 4 layers, identifying 3 layers as the optimal architecture. While a consistent gap between the 99% training F1 and the 90-92% test F1 still indicates some overfitting, the model's ability to generalize to unseen data is substantially better than all previous from-scratch models.

Part (f)

Table 14: Comparison of 'From Scratch' vs. 'Transfer Learning' Models

Model Type	Set	Precision	Recall	F1-Score
Scratch	Train	0.9998	0.9998	0.9998
Scratch	Test	0.9711	0.9710	0.9710
Transfer	Train	1.0000	1.0000	1.0000
Transfer	Test	0.9684	0.9683	0.9683

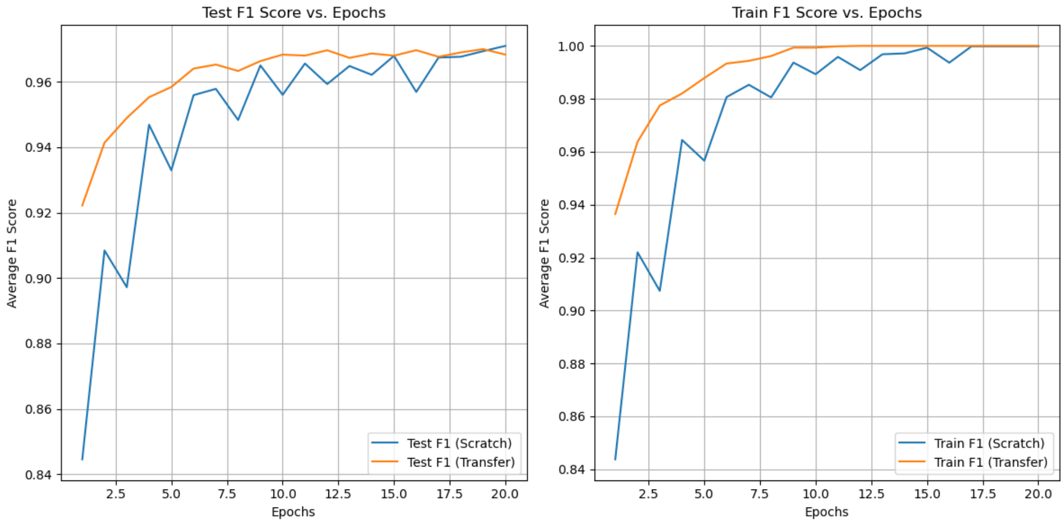

Figure 17: F1 Score vs Epoch for Train and Test sets

The transfer learning experiment demonstrates its primary benefit: **accelerated training**. The plots show the transfer model (orange line) starts with a higher F1 score on both the test and train sets, confirming it learned the digit features much faster by leveraging its pre-trained weights from the consonant dataset. Both models achieved outstanding final performance, indicating the digit task is much simpler. Interestingly, the model trained "from scratch" (blue line) eventually caught up and achieved a final Test F1 score of 97.1%, negligibly outperforming the transfer model's 96.8%. This suggests that while pre-training gave a powerful head start, the consonant features weren't perfectly optimal for the digit task, allowing the scratch model to find a slightly better-specialized solution in the end.